

Literature Review, Framework Landscape, Data Taxonomy & AI Research Primer

Assessing v7 against the published literature, mapping alternative frameworks considered and set aside, cataloguing the data sources shaping this field, and surveying where AI can extend transit-behavior modeling

Project: Evaluating Emerging AV Congestion, Safety, and Modal Shift Dynamics Using SFCTA's SF-CHAMP and Dynamic Network Simulation Framework
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Scope of this report: Literature review of v7's six methodological pillars; comparative framework landscape (why alternatives were not selected); public/internal/commercial data source inventory with leading organizations and researchers; a taxonomy primer for transportation dataset granularity; and an AI research frontier survey with case studies

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Headline finding. Every individual technique in v7 (nested logit with transferred coefficients, deadheading VMT strata, PopulationSim/IPU synthesis, Bayesian sequential safety monitoring, HCM incident-capacity modeling) has direct precedent in the peer-reviewed or agency literature — none of it is methodologically fringe. The most consequential gap this review surfaces is not statistical but architectural: SFCTA and SFMTA already collect and publish agency-authoritative data — an approved AV Permitting Framework with real disruption counts, Muni operator incident logs, and an INRIX-based congestion feed — that directly measures several of the same phenomena v7 is instead inferring from public survey microdata and simulated skims. That data is not in v7's ingestion layer today.

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1. Executive Summary

This report extends the earlier v6→v8 spec review into a full literature and landscape assessment. It asks three questions of v7's design: (1) does published research support each technique, (2) what alternative frameworks exist for the same problem and why weren't they chosen, and (3) what data sources — beyond what v7 currently ingests — are actually shaping this field, at SFCTA's own doorstep and nationally.

The review is organized around v7's six methodological pillars: mode-choice transferability, deadheading VMT estimation, the AV-transit-delay causal proxy, PopulationSim/IPU synthetic population, the Bayesian SPRT safety engine, and HCM incident-capacity modeling. For each, Section 2 summarizes what the literature supports and where it diverges from v7's implementation. Section 3 catalogs adjacent frameworks — other activity-based model platforms, other causal-inference tools, other safety-monitoring methods, other data-engineering stacks — and states plainly why each is not the right fit for a county-scale, SFCTA-context AV policy project. Sections 4–5 map the data and institutional landscape, including a specific finding that SFCTA's own Congestion Management Program dashboard already runs on INRIX probe data that v7 does not use. Section 6 is a standalone primer on transportation dataset taxonomy — telemetry, survey, spatial, administrative, operational, synthetic, and passive-device data — with native granularity versus typical analysis rollup for each. Section 7 surveys where AI research is actively extending transit-behavior modeling beyond what v7 attempts, with case studies.

2. Literature Review by Methodological Pillar

2.1 Nested Logit Mode Choice with Transferred AV Coefficients

No dataset available to this project — NHTS, CHTS, or BATS — has ever recorded a real person choosing an autonomous vehicle, since all three predate SF's commercial AV deployment. This is not unique to v7: it is the standard condition every AV mode-choice study has faced, and the field has a standard workaround. TUM's Department of Mobility Systems Engineering models AV modes using an *incremental logit* approach, treating a private AV as an improvement on driving alone and a shared AV as an improvement on bus transit, precisely because so little is directly known about AV-specific attributes. v7's "Mode Definition Drift" fix — probabilistically reassigning generic For-Hire trips to AV versus human-driven TNC — is structurally the same move.

Where the literature is less forgiving is on *which* parameters survive that transfer. A consistent finding across multiple transferability studies (Koppelman & Wilmot 1982/1986; Atherton & Ben-Akiva 1976; Sanko & Morikawa 2010) is that alternative-specific constants (ASCs) are the least stable, least transferable component of any borrowed discrete-choice model — time and cost sensitivities transfer reasonably well, but the ASC, which absorbs unobserved preference (comfort, novelty, trust), does not. A 2024 spatial-transferability study of shared electric-mobility hubs found that only a scaled model with an *updated* ASC meaningfully outperformed a naive transfer. v7 keeps its AV ASC (`b_0_AV`) as a flat assumed constant with no update mechanism — this is the single largest, most literature-confirmed source of downstream error in the pipeline.

2.2 Deadheading VMT Estimation

This is the best-precedented component of the project. Independent studies converge on a real range: national estimates of TNC deadheading run roughly 28–50% of total ride-hailing mileage (Henao & Marshall 2019; Wenzel et al. 2019; Komanduri et al. 2018), with an Austin-specific measurement putting it at 45%, nearly equal to loaded VMT. Critically, San Francisco already has its own published estimate: Cooper et al.'s direct measurement via the Uber/Lyft APIs, cited in a widely-referenced Science Advances congestion study, puts SF-specific deadheading at approximately 20% — versus roughly 50% in New York.

USABLE BENCHMARK

This SF-specific ~20% figure gives QA Gate 3 (Proxy 1 MAPE $\leq$ 12% vs. CPUC baseline) an external sanity check the current spec doesn't use: if a snapshot's computed deadheading ratio falls far outside the ~20–50% range this literature establishes, that is worth investigating before the number is trusted, independent of whether it clears the CPUC MAPE gate.

2.3 Causal Proxy 2 — AV-Induced Transit Delay

This is the review's most consequential finding. SFMTA has already run a version of this exact study using its own operator-reported incident logs: Muni drivers logged 12 driverless-vehicle disruption reports between September 2022 and March 2023, totaling at least 83 minutes of direct rider delay, and a separate SFMTA count found 92 unplanned AV stops from May–December 2022, 88% of them on transit-served streets. More recently, SFCTA's own September 2025 board-approved Conceptual Safety-Focused AV Permitting Framework Report contains updated figures — one operator alone logged nearly 20,000 unplanned stops of 2+ minutes, 62 stops of 15+ minutes, 17 vehicle-retrieval events, and 33 instances of obstructing first responders.

INGESTION GAP

v7's Causal Proxy 2 infers AV-induced transit delay from GTFS-RT breadcrumb deviations against a pre-incident baseline — an inferential proxy for something SFMTA and SFCTA already measure directly through agency incident logs and board-reported metrics. Neither source is in v7's ingestion inventory. This data should replace or, at minimum, validate the GTFS-RT-derived proxy and serve as an additional QA Gate 4 ground truth.

The same recent reporting also strengthens the placebo-independence caveat raised in the prior implementation review: large, spatially-correlated AV disruption events have occurred recently around San Francisco's July 4th fireworks crowds and a citywide December 2025 power outage, both producing simultaneous, non-independent stalls across multiple corridors — exactly the kind of correlated shock that violates the independence assumption behind a synthetic-control placebo test.

2.4 PopulationSim / Iterative Proportional Updating

PopulationSim's own documentation confirms v7's Gate 6 design is broadly right in spirit — the tool natively reports three diagnostics (average percent difference, its standard deviation, and percent RMSE) rather than a single MAPE figure, which is a refinement worth adopting. But a practitioner write-up from an MTC modeler documents a genuine gaming risk: with a single household-count control weighted at low importance, the tool validates poorly; raising that importance weight to 1000 makes it validate almost perfectly, regardless of whether the underlying fit actually improved. v7's Gate 6 has no check on the importance-weight configuration itself, meaning a "passing" gate could reflect a configuration choice rather than genuine representativeness.

2.5 Bayesian SPRT Safety Engine

Sequential Bayesian safety monitoring has real precedent — in a different domain than transportation. Risk-adjusted sequential probability ratio testing (RA-SPRT) was developed and validated in interventional cardiology and clinical device surveillance (Spiegelhalter et al.; Resnic et al.; Matheny et al.), where it was shown to detect true outlier institutions with a low false-positive rate when benchmarked against two multicenter clinical trials. That is a reassuring precedent for the statistical design of Gate 5's rare-event, low-VMT stress test. It also means this is a cross-domain transfer (clinical device monitoring → AV safety), which strengthens the case, raised in the prior implementation review, that v8 dropping this engine entirely — without flagging the removal — discards the single most rigorously-precedented component in the whole v6/v7 design. Separately, a 2022 study built a structured dataset of 260 California AV crashes (2019–2021) from DMV collision reports and applied Bayesian analysis to crash narratives — direct evidence that a Bayesian approach to AV-specific California crash data is both feasible and already precedented in the academic literature.

2.6 HCM Non-Linear Incident Capacity Drop

v7's rubberneck-factor bound (`F_rubberneck` $\in$ [0.05, 0.25]) is broadly consistent with published ranges. FHWA incident-management research documents rubbernecking-driven free-flow-speed reductions of roughly 5–10%, with one-lane-closure incidents separately reducing capacity by 17–30% depending on facility width — the lower bound of v7's range aligns well. The upper bound may be conservative in some cases: Dutch freeway research (Knoop et al. 2009) found capacity reductions from rubbernecking in the *opposing* or adjacent lanes were sometimes just as significant as

reductions in the incident's own direction — suggesting rubbernecking is not always the smaller, secondary term v_7 's additive formula treats it as.

3. Alternative Frameworks Considered — Fit Assessment

For each of v7's core methodological choices, this section names the leading alternative(s), what they are ideally suited for, and why they were not selected for this project's SF/SFCTA context.

3.1 Activity-Based Travel Demand Platforms

Framework	Ideal use case	Why not a fit here
MATSim	Large-scale, multi-agent co-evolutionary microsimulation; strong where dynamic traffic assignment and demand must be jointly re-optimized over many iterations (e.g., congestion-pricing studies, mobile-phone-record-fed regional models)	No existing MTC/SFCTA calibration lineage; adopting it means re-deriving a Bay-Area-calibrated model from scratch rather than building on SF-CHAMP, and its Java/Matsim-XML ecosystem doesn't interoperate cleanly with the Python-based ActivitySim stack SFCTA already runs
POLARIS	Fully integrated agent-persistence framework where agents and network simulation share one continuous state — well suited to DOE-funded, corridor-scale energy/resilience studies	Smaller user/support ecosystem outside its national-lab origin; no Bay Area implementation to build from
CT-RAMP (Java)	The predecessor architecture to ActivitySim — six-module microsimulation with intra-household interaction modeling	Directly superseded: ActivitySim <i>is</i> MTC's own CT-RAMP model, converted to Python starting in 2019, with PopulationSim integrated as its population-synthesis companion — this is already the region's adopted lineage, not an alternative to it
SimMobility / TRANSIMS	Stronger than MATSim/POLARIS specifically for long-term (land-use/demographic) simulation horizons	Project scope is a snapshot/near-term policy evaluation, not a long-horizon land-use forecast — the added complexity has no corresponding payoff here

WHY THIS MATTERS FOR FIT, NOT JUST PREFERENCE

ActivitySim's lineage as MTC's own converted regional model is the strongest argument for keeping it: any AV policy finding produced outside that lineage would not be directly comparable to the region's adopted travel model, undermining the project's usefulness to SFCTA and MTC decision-making.

3.2 Synthetic Population Generation

Framework	Ideal use case	Why not a fit here
PopulationSim / IPU (v7's choice)	Multi-level (household + person) marginal-control fitting with an established validation protocol; the ActivitySim-consortium standard	—
Deep generative synthesis (GAN/VAE-based population synthesizers)	Emerging research area; useful where marginal-control data is sparse or where higher-order joint correlations beyond what IPU controls for need to be preserved	Immature relative to IPU for agency production use — no established validation protocol equivalent to PopulationSim's percent-RMSE/percent-difference reporting, and no Bay Area precedent; the added modeling risk isn't justified when ACS/TAZ marginal controls are already available and sufficient
Simple IPF (no person-level joint)	Fast, simple, sufficient when only household-level controls matter	Cannot jointly satisfy household- <i>and</i> person-level controls simultaneously — a real limitation for this project, which needs person-level income tied to household composition for the EPC-aligned VOT segmentation

3.3 Causal Inference for the Transit-Delay Proxy

Framework	Ideal use case	Why not (yet) used here
21-day pre/post baseline (v7's choice)	Fast, simple, defensible when day-of-week effects are controlled for	—
Classical Synthetic Control (Abadie, Diamond & Hainmueller 2010, 2015)	Estimating the effect of an infrequent, geographically-bounded intervention by constructing a weighted combination of untreated comparison units matched on pre-treatment trajectory — the standard for comparative case studies with one or few treated units	Requires a defined pool of untreated "donor" corridors and enough pre-treatment history per corridor to fit convex weights reliably — heavier data requirements than v7's snapshot design currently budgets for, though it is the natural upgrade path
CausalImpact / Bayesian Structural Time Series (Brodersen et al. 2015, Google)	Fully Bayesian counterfactual estimation that infers the time-evolution of an effect and incorporates seasonality/trend priors directly — well suited to continuously-observed corridors like GTFS-RT delay series	Not yet adopted; this is arguably a better-fitting upgrade to Proxy 2 than the current placebo test, since it's built for exactly this kind of single-treated-unit time-series setting and natively handles the seasonality the current design ignores
Augmented / Synthetic Difference-in-Differences (Ben-Michael et al. 2021; Arkhangelsky et al. 2021)	Combines synthetic control's unit-weighting with diff-in-diff's time-adjustment; more robust when pre-treatment fit is imperfect	Most methodologically robust option surveyed, but also the most complex to implement and explain to non-technical stakeholders — a reasonable Phase 2 upgrade once Proxy 2 has more snapshot history to work with

3.4 Safety Monitoring Frameworks

Framework	Ideal use case	Why not a fit here
Bayesian SPRT (v6/v7's choice)	Continuous, rolling monitoring of a rate against a moving evidentiary threshold — well suited to an operator whose fleet and VMT grow over time and needs a live demotion/advancement signal	—
Highway Safety Manual Empirical Bayes (EB) / Safety Performance Functions	The AASHTO-codified, state-DOT-standard method for network screening — combining a calibrated crash-prediction regression with observed crash history to produce shrinkage-corrected, regression-to-the-mean-adjusted site-level crash estimates	Built for fixed roadway <i>segments</i> with multi-year historical crash counts and calibrated Crash Modification Factors — not for a single evolving fleet operator with a short operating history and no local SPF calibration; it answers "which locations are dangerous," not "is this operator's fleet safe enough to advance," which is v7's actual question
Statistical Process Control (CUSUM / EWMA)	Detecting a shift in a process mean as quickly as possible with a bounded false-alarm rate — common in manufacturing and public-health surveillance	Frequentist alternative to the same monitoring problem; loses the ability to express results as a probability statement (e.g., "95% probability the AV rate exceeds 1.5× human") that the Bayesian framing gives regulators and the public — a communication cost that outweighs CUSUM's simplicity for a policy-facing deliverable

3.5 Commercial Big-Data Mobility Platforms

Framework	Ideal use case	Why not a fit here
StreetLight Data	On-demand OD matrices, trip-purpose inference, and multimodal analytics built from location-based-services and connected-vehicle data — strong self-serve visualization layer	Subscription-licensed; not currently in v7's source inventory. Given SFCTA already licenses INRIX for its own congestion dashboard (see §4), adding a second commercial probe-data vendor is a procurement question, not a methodological one
Replica	Synthetic, privacy-preserving population-and-trip datasets built from a blend of cellular, LBS, and survey data — increasingly used as a research substrate (e.g., in Graph-RAG-augmented LLM mode-choice studies, §7)	A private synthetic population is a plausible substitute for, or supplement to, v7's own PopulationSim-built population — but introduces a third-party black-box population alongside the project's own, with reconciliation overhead not currently budgeted
INRIX	Real-time and historical probe-based speed/travel-time data aggregated from connected vehicles, fleets, and mobile devices — the current input to SFCTA's own public Congestion Dashboard	Already licensed by SFCTA operationally, but not wired into v7's ingestion layer, which instead relies on OSMnx static topology plus DTA-simulated skims for travel time. See the specific gap noted in §4.2.

3.6 Large-Scale Geospatial Data Engineering

Framework	Ideal use case	Why not a fit here
geopandas / Shapely (v7's choice)	In-memory vector GIS operations at city-county scale — sufficient for SF's data volumes	—
Apache Sedona / GeoSpark, Databricks geospatial	Distributed spatial joins and clustering at true big-data scale (statewide/national probe-data volumes, billions of pings)	Overkill at San Francisco's single-county scale — would add cluster infrastructure and operational overhead with no corresponding performance need; worth revisiting only if the project scope expands to regional (nine-county Bay Area) analysis

4. Data Source Landscape — Public, Internal & Commercial

4.1 Public / Federal & State

Source	What it provides	Steward
NHTS	National household travel diary survey; mode, time, purpose microdata	FHWA / Oak Ridge National Laboratory
CHTS	California-specific household travel survey (2010–2013 vintage)	Caltrans
ACS / ACS PUMS	Household/person demographic microdata and TAZ-level marginal controls	U.S. Census Bureau
LEHD LODES	Home-to-work commute OD flows, block-level	U.S. Census Bureau
NHTSA SGO	Federally-reportable ADS/L2 crash incident reports	NHTSA
CA DMV AV Collision Reports	California-specific AV crash reports, used as the basis for academic Bayesian crash-narrative research	California DMV
PeMS	Freeway loop-detector volumes, speeds, occupancy — statewide, high-frequency	Caltrans / UC Berkeley
CPUC Quarterly Filings	Commercial passenger-service VMT — v7's designated denominator source of truth	California Public Utilities Commission
GTFS / GTFS-RT	Static transit schedules and real-time vehicle position/delay feeds	SFMTA, published via 511.org
OpenStreetMap	Road network topology and edge attributes	OSM Foundation / community-maintained

4.2 Regional & Internal (SFCTA / SFMTA / MTC)

Source	What it provides	In v7 ingestion?
BATS (Bay Area Travel Study)	MTC household travel survey; app-based (rMove) GPS-assisted trip and day files, income-linked	Yes
SFCTA Congestion Dashboard	Monthly automobile Level-of-Service and VMT per corridor, built on INRIX Roadway Analytics speed data	No — v7 relies on DTA-simulated skims instead of this already-licensed observed-speed feed
SFMTA Muni Operator Incident Logs	Driver-reported AV disruption events with direct delay-minute counts	No
SFCTA AV Permitting Framework Report (Sept. 2025)	Board-approved five-stage performance-based AV permitting framework; published unplanned-stop counts, collision figures, first-responder obstruction events	No
SFCTA "TNCs 2020" Report	Statewide ride-hailing activity profile built from CPUC annual filings — precedent for CPUC-as-source-of-truth	Precedent only; not directly ingested
SF-CHAMP	SFCTA's own regional travel demand model — the DTA/skim engine v7 iterates against	Yes — core to Gate 2

REPEATED FINDING, STATED ONCE MORE DIRECTLY

Three of the five internal SFCTA/SFMTA sources above are not in v7's ingestion layer, despite being the most directly authoritative data available to a project run under SFCTA's own name.

4.3 Commercial

StreetLight Data, Replica, and INRIX are the three leading commercial mobility-analytics vendors serving state DOTs and MPOs (e.g., Arizona DOT and Ohio DOT both run INRIX-plus-StreetLight stacks; TxDOT holds both INRIX and Replica subscriptions). See §3.5 for fit assessment.

5. Leading Organizations & Researchers

Organization / Group	Relevance to this project
UC Berkeley SafeTREC & Institute of Transportation Studies (ITS)	California's leading transportation safety research center; steward of much of the crash and safety-methodology literature cited in §2.5
UC Berkeley PATH	Long-running program on automated/connected vehicle research and testing methodology
MTC / RSG	Stewards of ActivitySim and PopulationSim — the platforms v7's PopulationSim integration builds on; RSG converted MTC's CT-RAMP model into the current open-source codebase
SFCTA / SFMTA	The project's own home agencies; source of the AV Permitting Framework Report and Muni incident logs identified in §4.2 as an ingestion gap
TRB (Transportation Research Board)	Standards body for nested-logit mode-choice practice, HCM incident-capacity methodology, and AV policy committees
Google Research (Brodersen, Gallusser, Koehler, Remy, Scott)	Authors of the CausallImpact / Bayesian structural time-series method identified in §3.3 as a plausible upgrade to Proxy 2

Researcher	Known for / relevance here
Frank Koppelman & C. G. Wilmot	Foundational discrete-choice model transferability analysis (1982, 1986) — the ASC-instability finding underlying §2.1
Moshe Ben-Akiva	Co-developer of the nested logit / random utility framework underlying all of v7's mode-choice work
Michel Bierlaire	Developer of Biogeme, the open-source MLE estimation tool used across the AV mode-choice literature reviewed in §2.1
Ezra Hauer	Originator of the Empirical Bayes safety method underlying the Highway Safety Manual, discussed in §3.4
Alberto Abadie	Originator of the classical Synthetic Control method (with Diamond & Hainmueller), discussed in §3.3
Susan Shaheen	UC Berkeley Transportation Sustainability Research Center; leading shared-mobility and TNC-impact researcher
Joan Walker	UC Berkeley; discrete-choice and activity-based modeling, including AV-inclusive mode choice specification work
Giovanni Circella	UC Davis 3 Revolutions Future Mobility program; travel-behavior and multitasking-in-AVs research cited in §2.1's mode-choice literature
Kara Kockelman	UT Austin; widely cited AV/TNC deadheading and VMT-impact research

6. Taxonomy of Transportation Datasets — A Working Primer

This section is a standalone reference: it classifies transportation data by *type*, states the *native granularity* at which it is captured, and the *typical rollup level* at which it is actually consumed for planning-grade analysis — the gap between the two is usually where implementation errors happen.

Category	Examples	Native granularity	Typical analysis rollup	Notes
Telemetry / GPS & probe	AV onboard telemetry, TNC trip pings, GTFS-RT, INRIX/StreetLight probe data, cellular CDR	Sub-second to few-second location fixes, per device/vehicle	Corridor-level speed/travel-time, hourly or 15-minute bins, monthly trend	Highest resolution but noisiest; requires anomaly filtering (Rule 3, §4.3 of the implementation spec) before any rollup
Household travel survey	NHTS, CHTS, BATS	Individual trip/day diary per person, self-reported timestamps, home-geocoded (PUMA/Census tract, sometimes finer under restricted access)	PUMA or county-level mode share; TAZ-level with a confidential geography extension	Ground-truth for revealed behavior, but infrequent (multi-year vintage) and subject to recall/reporting bias
Spatial / network	OpenStreetMap, SFCTA TAZ shapefile, HIN/EPC layers	Edge/node or parcel-level geometry	TAZ, corridor, or Census Block Group for most planning outputs	Rarely updated relative to telemetry — treat as a slowly-changing dimension
Administrative / regulatory filings	CPUC quarterly VMT filings, DMV testing-VMT filings, NHTSA SGO	Operator-level, quarterly or per-incident	Operator-level or citywide quarterly aggregate	Authoritative but low-frequency; not a substitute for continuous monitoring
Operational / agency	SFMTA Muni operator incident logs, AVL/APC transit data, signal timing logs	Per-incident or per-vehicle-run	Route-level or citywide monthly summary	Directly authoritative for agency-specific outcomes (delay minutes, obstruction events) — see §4.2 gap
Synthetic / derived	PopulationSim output, DTA-simulated skims, IPF-balanced OD matrices	Synthetic household/person or OD-cell level, matching the control geography used to build it	TAZ-to-TAZ or income-tier aggregate	Only as reliable as its calibration inputs — always trace back to the underlying seed/control data before treating as ground truth
Crowd-sourced / passive-device	Waze incident reports, SafeGraph/Advan place-visit data, commercial LBS panels (Replica, StreetLight)	Device-ping or user-report level, often sub-minute	Zone-level daily/hourly visit counts or OD matrices	Panel is not a random sample of the population — device/app penetration varies sharply by demographic and geography

6.1 Reading the Taxonomy

- **Granularity almost always exceeds the level at which decisions are made.** Second-by-second GPS pings are rolled up to hourly corridor speeds; individual trip diaries are rolled up to TAZ mode shares. The rollup step is where information is lost and where validation (Gates 3, 4, 6 in v7) has to do its work.
- **Survey and administrative data are slow-moving; telemetry and crowd-sourced data are fast-moving.** Combining them (as v7 does throughout) requires an explicit temporal-alignment decision — this is exactly what the "Information Gaps" caveat about survey/telemetry vintage mismatch (§2.1, §10 of the earlier implementation review) is about.
- **"Public" and "restricted" are different products from the same survey.** BATS/NHTS/CHTS public releases are geography-coarsened for privacy; the TAZ-level detail this project needs requires a data-sharing agreement, not just a public download (see the earlier discussion of BATS restricted geography).

7. AI Research Frontiers for Transit Behavior Modeling

This section surveys active research areas where AI methods are being used to capture transit behavior and predictors beyond what v7's nested-logit-plus-IPU approach attempts — with case studies where available.

7.1 LLM-Based Generative Agents for Activity Simulation

A fast-growing 2024–2026 research line replaces rule-based activity-generation logic (the core of ActivitySim's tour-scheduling modules) with LLM-driven agents that plan, reflect, and adapt.

CASE STUDY — GATSIM (2025)

Generative-Agent Transport Simulation equips agents with socioeconomic profiles, psychologically-informed memory, and tool use, letting them plan and adapt daily travel rather than sampling from fixed choice probabilities. In blind evaluation against human annotators across 50 role-played mobility scenarios, an independent LLM judge rated the generative agents as more realistic than the human-authored scenarios in 46% of cases (versus 40% preferring the human version) — a rough parity result that is notable given the agents received no explicit behavioral rule-coding.

CASE STUDY — COGNITIVE AGENTS IN URBAN MOBILITY (2025)

LLM-based agents with multi-horizon memory were integrated into the SimFleet agent-based simulator across a 20-day simulation of 320+ individuals in Valencia, Spain, including a scenario with only 40% taxi availability. Agents showed stable strategic adaptation to the disruption while retaining flexibility to fall back to alternate modes — directly relevant to modeling how SF travelers might adapt if AV or transit availability shifts under a policy change.

CASE STUDY — GRAPH RAG "PREFERENCE CHAIN" (2025)

Combines Graph Retrieval-Augmented Generation with an LLM to simulate context-aware mode-choice behavior, tested directly against the **Replica** commercial mobility dataset (§4.3) and shown to outperform a standard LLM baseline at matching real-world mode-choice patterns — a concrete example of the commercial-dataset-plus-LLM combination discussed as a possible v7 upgrade path in §3.5.

Relevance to v7: this line of research is the natural extension of the ASC-transferability problem in §2.1 — instead of assuming a fixed AV-specific constant, an LLM agent with memory can in principle adapt its "attitude" toward AVs over simulated exposure, which is closer to how real preference formation for a genuinely new mode is believed to work.

7.2 Graph Neural Networks for OD Demand Prediction

CASE STUDY — MULTI-GRAPH CONVOLUTIONAL NETWORKS FOR RIDE-SOURCING DEMAND (KE ET AL., 2021, TRANSPORTATION RESEARCH PART C)

A spatio-temporal encoder-decoder residual multi-graph convolutional network was built specifically to predict ride-sourcing OD demand, jointly modeling multiple graph structures (spatial adjacency, functional similarity, and network connectivity) rather than a single distance-based graph. This is directly applicable to v8's static IPF-based OD balancing against LEHD LODES (which is commute-only and updates with a multi-year lag) — a GNN-based approach could produce a continuously-updated, ride-sourcing-specific OD estimate rather than relying on a stale, work-trip-only administrative dataset.

Broader relevance: GNN-based traffic and OD forecasting is now a mature enough subfield to have its own literature-tracking repositories (e.g., the GNN4Traffic collection), with applications spanning bike-share, metro, and ride-hailing demand — all structurally closer to v7/v8's OD-flow problem than the static-IPF approach currently specified.

7.3 NLP on Crash Narratives

The 2022 study on 260 California AV crashes (§2.5) used text analytics on crash narratives combined with Bayesian analysis to structure otherwise free-text DMV collision reports into an analyzable safety dataset. This is a direct, low-lift extension for this project: NHTSA SGO and CA DMV filings both include free-text incident descriptions that v7's ingestion layer currently discards in favor of structured fields only (VIN, timestamp, injury severity) — an NLP extraction layer could recover causal-factor detail (e.g., distinguishing a stalled-vehicle blockage from a collision from a first-responder-obstruction event) without waiting for a schema change from NHTSA or DMV.

7.4 Causal Machine Learning

Beyond the synthetic-control-family methods in §3.3, causal ML techniques (double machine learning, causal forests) are increasingly used to estimate heterogeneous treatment effects — e.g., whether AV-induced transit delay differs systematically by neighborhood income tier or EPC status, rather than producing one citywide average effect. This would let Causal Proxy 2's output speak directly to the project's equity mandate in a way the current single-point delay estimate cannot.

7.5 Reinforcement Learning for Fleet Rebalancing

A separate research thread applies RL to AV/TNC fleet dispatch and rebalancing — directly relevant to the deadheading VMT problem in §2.2, since deadheading is, in large part, an artifact of how a fleet operator chooses to reposition idle vehicles. While not part of v7's scope (v7 measures deadheading, it doesn't model operator dispatch decisions), this is the natural research frontier if a future phase of the project moves from measuring deadheading to informing policy that could influence it (e.g., staging-zone requirements tied to permit tier).

8. Synthesis & Recommendations

1. **Ingest SFCTA's own AV Permitting Framework Report and SFMTA Muni incident logs.** This is the single highest-value fix identified across both the implementation review and this literature review — it replaces or validates an inferential proxy with agency-authoritative ground truth that already exists.
2. **Wire SFCTA's existing INRIX feed into the skim-validation step.** QA Gate 2 currently validates DTA-simulated skims only against each other (RMSE convergence); validating against observed INRIX speeds SFCTA already licenses would be a stronger check with no new procurement.
3. **Treat CausallImpact/Bayesian structural time series as the near-term upgrade path for Proxy 2,** ahead of full synthetic control or synthetic diff-in-diff — it fits the continuously-observed GTFS-RT data v7 already collects and directly addresses the seasonality gap noted in the earlier methodology caveats.
4. **Restore or replace the Bayesian SPRT safety engine before treating v8 as production-ready** — it is the most rigorously precededented component in the whole design (per the clinical RA-SPRT literature in §2.5), and its removal in v8 was not disclosed as a trade-off.
5. **Add an NLP layer over free-text crash-narrative fields** in NHTSA SGO and CA DMV filings — a low-cost way to recover causal detail the structured fields currently discard.
6. **Treat GNN-based OD prediction as a medium-term alternative to static IPF/LODES balancing** for ride-sourcing-specific flows, given LODES' commute-only scope and multi-year publication lag.

References

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